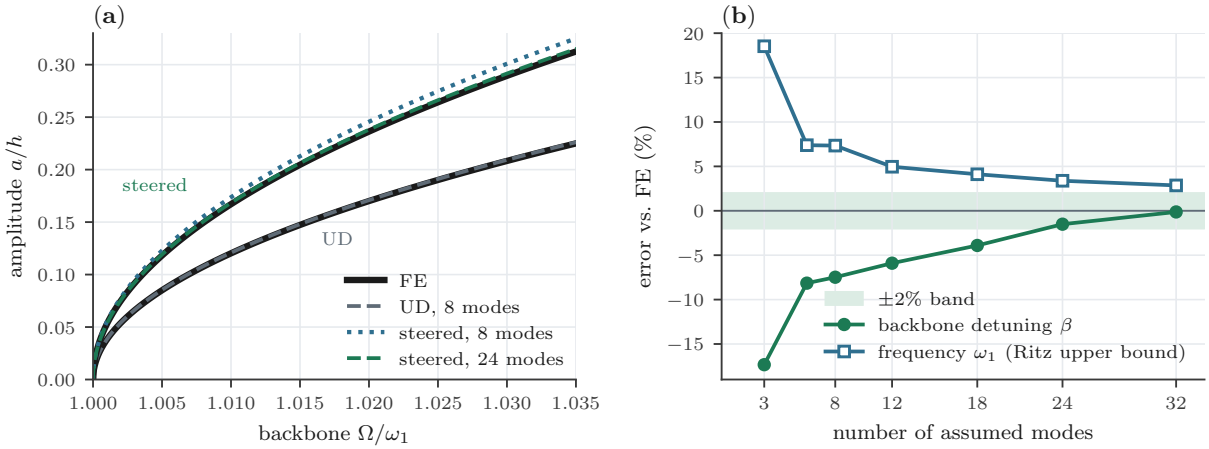

(a)
amplitude a/h
0.00
0.05
0.10
0.15
0.20
0.25
0.30
steered
UD
FE
UD, 8 modes
steered, 8 modes
steered, 24 modes
1.000
1.005
1.010
1.015
1.020
1.025
1.030
1.035
backbone Ω/ω_1
(b)
error vs. FE (%)
20
15
10
5
0
−5
−10
−15
±2% band
backbone detuning β
frequency ω_1 (Ritz upper bound)
3
8
12
18
24
32
number of assumed modes